



Amir Adam

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● ABOUT ME

Embedded Systems and IoT engineering student with hands-on experience in firmware development using C/C++ and Python on ESP32, STM32, and Raspberry Pi. Strong in communication protocols, system integration, and building end-to-end solutions for sensor interfacing, data acquisition, and network communication. Seeking to apply practical problem-solving and debugging skills in real-world engineering environments.

● EDUCATION AND TRAINING

16/08/2022 – CURRENT Mohali , India

BACHELOR OF ENGINEERING IN ELECTRONICS & COMMUNICATIONS Chandigarh University

Relevant Courses: Embedded Systems | Communications | VLSI Design

Level in EQF EQF level 6

13/02/2017 – 01/12/2020 Wau, South Sudan

CERTIFICATE OF SECONDARY EDUCATION St.Ignatius of Loyola Secondary School

General Sciences: Maths | Physics | Chemistry | Biology

Level in EQF EQF level 4

● SKILLS

Programming

C/C++ | Python | Typescript

Embedded Systems

Microcontrollers: ESP32, STM32 | Raspberry Pi | Sensor Integration

Tools & Platforms

Git | STM32CUBEIDE | VSCode | GitHub | Arduino IDE / PlatformIO | Github

App Development

React Native | Supabase | Expo/Expo router | PostgreSQL

Conceptual Knowledge

Communication Protocols: I2C, SPI, UART | Embedded Systems Architecture | Networking Basics (HTTP, client-server side) | Basic PCB Design

● LANGUAGE SKILLS

Mother tongue(s): **ARABIC(SUDANESE DIALECT)**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **PROJECTS**

15/03/2026 – 20/04/2026

IoT-Based Food Monitoring System

- Tech Stack:** ESP32, AWS IoT Core, AWS Lambda, Firebase (Firestore, FCM), React Native, MQTT, Wi-Fi
- Designed and implemented an **end-to-end IoT architecture** for real-time food freshness monitoring using ESP32 and environmental sensors (DHT11, MQ135).
 - Engineered **secure MQTT communication over TLS** to stream sensor data to AWS IoT Core with reliable cloud ingestion.
 - Developed **serverless backend logic using AWS Lambda** to process, validate, and compute food freshness metrics from sensor data.
 - Integrated **Firebase Firestore** for real-time data synchronization and historical storage, enabling low-latency dashboard updates.
 - Implemented **push notification system using Firebase Cloud Messaging (FCM)** to alert users on food spoilage conditions.
 - Built a **React Native mobile application** with real-time dashboard visualization and device interaction.
 - Architected a **scalable, event-driven pipeline** combining AWS and Firebase services for efficient data flow and system responsiveness.
 - Successfully tested system with real-world samples, validating sensor accuracy and end-to-end reliability.

20/10/2025 – 20/11/2025

Disaster-Resilient Offline Mesh Communication System

Disaster-Resilient Offline Mesh Communication System

Technologies: ESP32, LoRa

- Designed and developed an **offline mesh communication system** using **ESP32** and **LoRa** to enable connectivity in network-isolated environments.
- Implemented a **peer-to-peer mesh networking architecture** for decentralized message transmission without cellular or internet dependency.
- Integrated **sensor data acquisition and local processing** to enhance situational awareness in emergency scenarios.
- Optimized embedded firmware for **low-power operation**, improving reliability in constrained and disaster-prone environments.

Link <https://techtacklenerd.blogspot.com/2025/11/disaster-resilient-hybrid-mesh.html>

10/03/2026 – 24/03/2026

Edge AI Surveillance System with Real-Time Intrusion Detection (Raspberry Pi 5)

Technologies: Python, OpenCV, TensorFlow Lite, SQLite, FastAPI, Telegram Bot API

- Designed and implemented a **real-time edge AI surveillance system** on Raspberry Pi 5 for human detection using a **TensorFlow Lite MobileNet SSD model**.
- Built a **modular processing pipeline** including video ingestion, preprocessing (**OpenCV**), AI inference, and spatial decision logic for intrusion detection.
- Developed a **custom spatial logic engine** to evaluate object position (center of mass and foot approximation) against defined restricted zones.
- Implemented **asynchronous alerting using multithreading**, delivering real-time intrusion notifications (image + metadata) via **Telegram Bot API**.

- Engineered a **local web interface using FastAPI** to stream annotated MJPEG video and visualize detection logs stored in **SQLite**.
- Integrated **stateful logging** for event tracking, including timestamps, confidence scores, and snapshot storage for post-event analysis.

Link <https://github.com/amirphiladam2/edge-ai-surveillance>

15/02/2026 – 19/02/2026

Full-Stack IoT Monitoring System with Web Dashboard

Technologies: ESP32, React (Vite), Python (Flask), HTTP, REST API

- Designed and developed a **full-stack IoT system** for real-time sensor data acquisition and visualization.
- Implemented firmware on **ESP32** to collect environmental sensor data and transmit it to a backend server via **HTTP requests**.
- Built a **RESTful API using Flask** to handle data ingestion, processing, and storage.
- Developed a **responsive frontend dashboard using React (Vite)** to display real-time and historical sensor data.
- Enabled **dynamic data synchronization** through seamless **API integration** between frontend and backend.
- Improved system reliability by handling **network latency**, retries, and ensuring consistent data transmission.

Link <https://github.com/amirphiladam2/FullStack-IoT>

● CREATIVE WORKS

12/07/2023 – CURRENT

YouTube Tech Content Creator

- Created technical educational content focused on Embedded Systems and App development concepts.
- Produced video tutorials demonstrating real-world implementations using STM32, ESP32, React Native.

Link <https://www.youtube.com/@amirdevstudios/videos>

● HOBBIES AND INTERESTS

Nature Photography

- Nature photography, with a focus on landscapes and environmental elements

Travelling

- Travelling and exploring new places and cultures